

torch.addcmul#

<https://pytorch.org/docs/stable/generated/torch.addcmul.html>

Description:

Performs the element-wise multiplication of tensor1 by tensor2, multiplies the result by the scalar value and adds it to input. The shapes of tensor, tensor1, and tensor2 must be broadcastable. For inputs of type FloatTensor or DoubleTensor, value must be a real number, otherwise an integer. input (Tensor) – the tensor to be added tensor1 (Tensor) – the tensor to be multiplied

Function Signature

```
torch.addcmul(input, tensor1, tensor2, *, value=1, out=None)  
→ Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

`value` (Number, optional) – multiplier for `tensor1.*tensor2`
`tensor1.*tensor2` out (Tensor, optional) – the output tensor.

Code Examples

```
>>> t = torch.randn(1, 3)
>>> t1 = torch.randn(3, 1)
>>> t2 = torch.randn(1, 3)
>>> torch.addcmul(t, t1, t2, value=0.1)
tensor([[ -0.8635, -0.6391,  1.6174],
        [-0.7617, -0.5879,  1.7388],
        [-0.8353, -0.6249,  1.6511]])
```

Detailed Documentation

Documentation

Performs the element-wise multiplication of tensor1 by tensor2, multiplies the result by the scalar value and adds it to input.

The shapes of tensor, tensor1, and tensor2 must be broadcastable.

For inputs of type FloatTensor or DoubleTensor, value must be a real number, otherwise an integer.

input (Tensor) – the tensor to be added

tensor1 (Tensor) – the tensor to be multiplied

tensor2 (Tensor) – the tensor to be multiplied

value (Number, optional) – multiplier for $\text{tensor1} \times \text{tensor2}$
 $\text{tensor1} \times \text{tensor2}$

out (Tensor, optional) – the output tensor.